NOTES: CLASS-5

Half Range Fourier series of f(x) in(0,π)

Cosine series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

Fourier Coefficients:

$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx.$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\, dx$$

Sine Series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

Fourier Coefficients:

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\, dx$$

Half Range Fourier series of f(x) in (0,l):

Cosine series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l}$$

Fourier Coefficients:

$$a_0 = \frac{2}{l}\int_0^{l} f(x)dx.$$

$$a_n = \frac{2}{l}\int_0^{l} f(x)\cos\frac{n\pi x}{l}\, dx$$

Sine Series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}$$

Fourier Coefficients:

$$b_n = \frac{2}{l}\int_0^{l} f(x)\sin\frac{n\pi x}{l}\, dx$$

Problem 1 . Obtain the Fourier expansion of the function $f(x) = xsinx,$ as a cosine series in $(0, \pi)$.

Solution : The Fourier cosine series is given by,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n cosnx$$

where

$$a_0 = \frac{2}{\pi}\int_0^{\pi} xsinxdx$$

$$= \frac{2}{\pi}[x(-cosx) - (1)(-sinx)]_0^{\pi}$$

$$= \frac{-2}{\pi}\{\pi\cos\pi - 0\}$$

$$\Rightarrow a_0 = 2$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} xsinxcosnxdx$$

$$= \frac{1}{\pi}\int_0^{\pi} x[sin(n+1)x - \sin(n-1)]dx$$

$$= \frac{1}{\pi}x.\left[-\frac{\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1} - (1).\left[-\frac{\sin(n+1)x}{n+1} + \frac{\sin(n-1)x}{n-1}\right]\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[-\frac{\pi(-1)^{n+1}}{n+1} + \pi\frac{(-1)^{n-1}}{n-1}\right]$$

$$= \frac{\pi(-1)^n}{\pi}\left[\frac{1}{n+1} - \frac{1}{n-1}\right]$$

$$= (-1)^n\left[\frac{-2}{n^2-1}\right]$$

$$\Rightarrow a_n = \frac{2(-1)^{n+1}}{n^2-1} \quad \text{where } n \neq 1$$

When $n = 1$,

$$a_1 = \frac{2}{\pi}\int_0^{\pi} xsinxcosxdx$$

$$= \frac{1}{\pi}\int_0^{\pi} xsin2xdx$$

$$= \frac{1}{\pi}\left[\frac{x(-cos2x)}{2} - \frac{(1)(-sin2x)}{4}\right]_0^{\pi}$$

$= \frac{1[-\pi]}{2\pi}$

$\Rightarrow a_n = \frac{1}{2}$

Therefore the Half range Fourier Cosine Series for $f(x)$ is

$$f(x) = x\,sinx = \frac{\square_0}{2} + a_1 cosx + \sum_{n=2}^{\infty} a_n cosnx$$

$$\Rightarrow \quad x\,sinx = 1 - \frac{cosx}{2} + \sum_{n=2}^{\infty} \frac{2(-1)^{n+1}}{n^2-1} cosnx$$

Problem 2. Find the Fourier half range a) Cosine series b) Sine series of

$$\mathbf{f}(\boldsymbol{x}) = \begin{cases} \boldsymbol{x}, & \mathbf{0} < \boldsymbol{x} < \mathbf{1} \\ \mathbf{2} - \boldsymbol{x}, & 1 < x < \mathbf{2} \end{cases}$$

Solution : The Half Range Fourier Cosine series is given by,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n cos\frac{n\pi x}{l}$$

where

$$a_0 = \frac{2}{2}\int_0^2 f(x)dx$$

$$= \int_0^1 x\,dx + \int_1^2 (2-x)\text{dx}$$

$$= \left[\frac{x^2}{2}\right]_0^1 + \left[2x - \frac{x^2}{2}\right]_1^2$$

$$= 1$$

$$a_n = \frac{2}{l}\int_0^l f(x)\,cos\frac{n\pi x}{l}\,dx$$

$$= \frac{2}{2}\int_0^2 f(x)cos\frac{n\pi x}{2}dx$$

$$= \int_0^1 x\cos\frac{n\pi x}{2}dx + \int_1^2 (2-x)\cos\frac{n\pi x}{2}dx$$

$$= \left[x.\ \frac{\sin\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right]_0^1 - \left[-\frac{\cos\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right]_0^1 + \left[(2-x).\ \frac{\sin\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right]_1^2 - \left[\frac{\cos\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right]_1^2$$

$$= \frac{2}{n\pi}\sin\left(\frac{n\pi}{2}\right) + \frac{4}{n^2\pi^2}\left[\cos\left(\frac{n\pi}{2}\right) - 1\right] - \frac{2}{n\pi}\sin\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2}\left[\cos n\pi - \cos\left(\frac{n\pi}{2}\right)\right]$$

$$= \frac{8}{n^2\pi^2}\cos\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2}[1+(-1)^n]$$

b) The Half range Fourier Sine series is given by, $f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}$

where $b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$

$$= \frac{2}{2}\int_0^2 f(x)\sin\frac{n\pi x}{2}dx$$

$$= \int_0^1 x\sin\frac{n\pi x}{2}dx + \int_1^2 (2-x)\sin\frac{n\pi x}{2}dx$$

$$= \left[x.\ \frac{-\cos\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right]_0^1 - \left[-\frac{\sin\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right]_0^1 + \left[(2-x).\ \frac{-\cos\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right]_1^2 - \left[\frac{\sin\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right]_1^2$$

$$= \frac{-2}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{4}{n^2\pi^2}\left[\sin\left(\frac{n\pi}{2}\right)\right] + \frac{2}{n\pi}\cos\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2}\left[0 - \sin\left(\frac{n\pi}{2}\right)\right]$$

$$= \frac{8}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)$$

Therefore,

i. Half range Cosine series: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{2}$

$$\Rightarrow f(x) = \frac{1}{2} + \sum_{n=1}^{\infty}\frac{8}{n^2\pi^2}\cos\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2}[1+(-1)^n]\cos\frac{n\pi x}{2}$$

ii. Half range Sine series : $f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}$

$$\Rightarrow f(x) = \sum_{n=1}^{\infty}\frac{8}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)\sin\frac{n\pi x}{2}$$

Problem 3. By using the sine series for $f(x) = 1$ in $0 < x < \pi$. Hence show that,

$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$.

Solution : The half range Fourier sine series given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$where\ b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\ sinnx\ dx$$

$$\Rightarrow b_n = \frac{2}{\pi}\int_0^{\pi}(1)\ sinnx\ dx$$

$$= \frac{2}{\pi}\left[\frac{-cosnx}{n}\right]_0^{\pi} = \frac{-2}{n\pi}[cosn\pi - 1]$$

$$= \frac{-2}{n\pi}[(-1)^n - 1]$$

$$= \frac{2}{n\pi}[1 - (-1)^n]$$

Therefore the Half range Fourier Sine series of $f(x)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}$$

$$\Rightarrow 1 = \sum_{n=1}^{\infty}\frac{2}{n\pi}[1 - (-1)^n]\sin nx$$

$$\Rightarrow 1 = \frac{4}{\pi}\left(sinx + \frac{sin3x}{3} + \frac{sin5x}{5} + \cdots.\right)$$

Deduction :

Consider the Parseval's identity for odd functions given by,

$$2\int_0^l [f(x)]^2 dx = l[\sum_{n=1}^{\infty} {b_n}^2]$$

$$\Rightarrow 2\int_0^{\pi}(1)^2 dx = \pi\left[\sum_{n=1}^{\infty}(\frac{\mathbf{2}}{\boldsymbol{n\pi}}[\mathbf{1} - (-\mathbf{1})^{\boldsymbol{n}}])^2\right]$$

$$\Rightarrow 2[x]_0^{\pi} = \frac{4}{\pi^2}\left[4 + \frac{4}{3^2} + \frac{4}{5^2} + \cdots \ldots \ldots \ldots \ldots \ldots \ldots ..\right]$$

$$\Rightarrow \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \ldots\ldots\ldots\ldots\ldots = \frac{\pi^2}{8}$$

REFERENCES:

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2. **Higher Engineering Mathematics-B.V. Ramana**

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